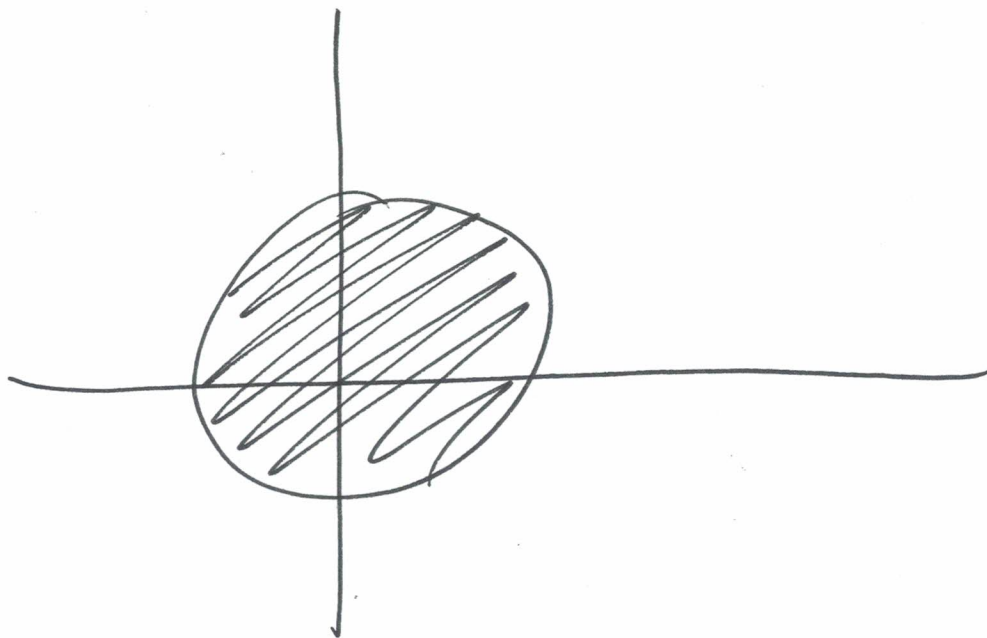


TWO RANDOM VARIABLES

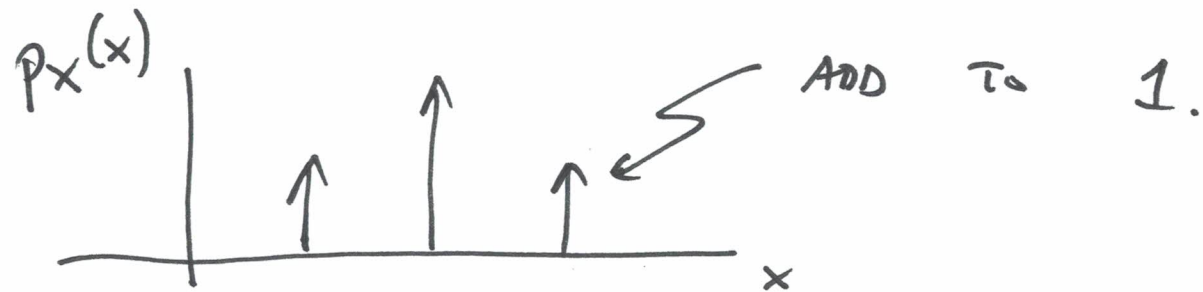
(X, Y) COULD BE DISCRETE
CONTINUOUS



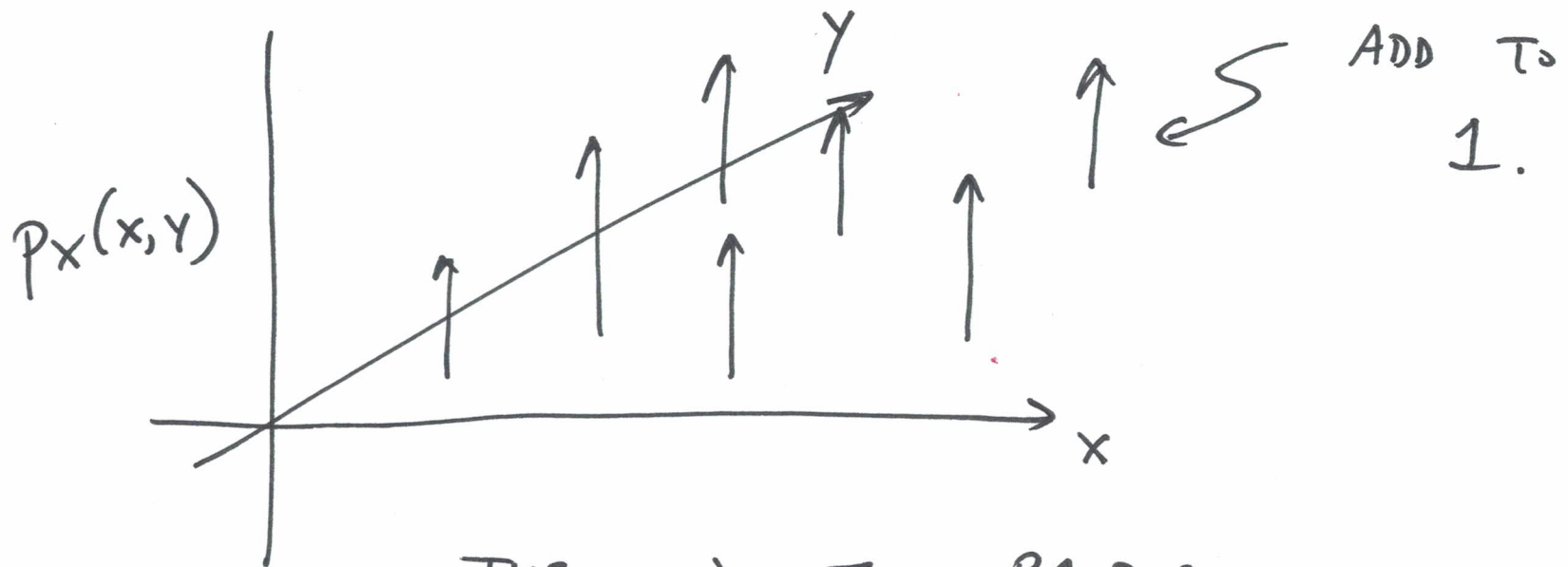
TODAY: DISCRETE. NEXT TIME: CONTINUOUS

DISCRETE 2D PMF / CDF

1-D :



2-D



THE JOINT PROBABILITY
MASS FUNCTION (PMF)

$$P_{XY}(x, y) = P(X = x, Y = y)$$

$\uparrow$
 NAMES
 OF
 RANDOM
 VARIABLES

$\uparrow$
 POSSIBLE
 OUTCOMES

$x \backslash y$	0	1	2
0	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
1	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
2	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

$P_{XY}(x, y)$

JOINT CDF:

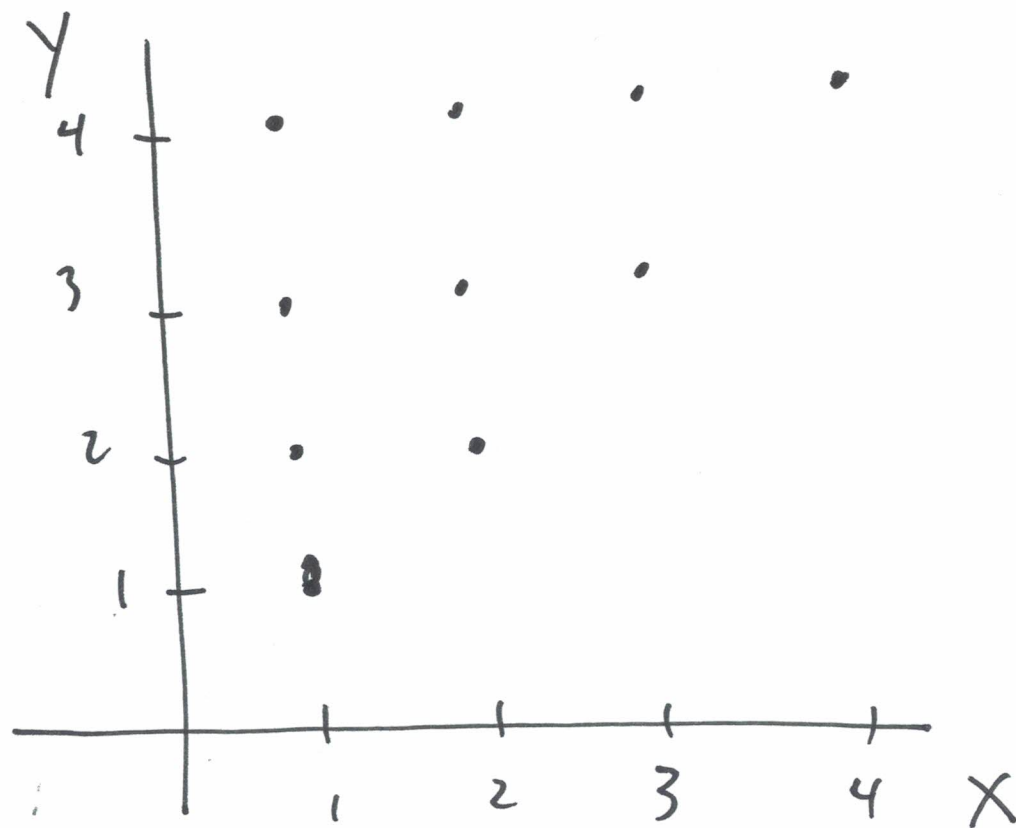


$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

$$P_{XY}(x,y) = C$$

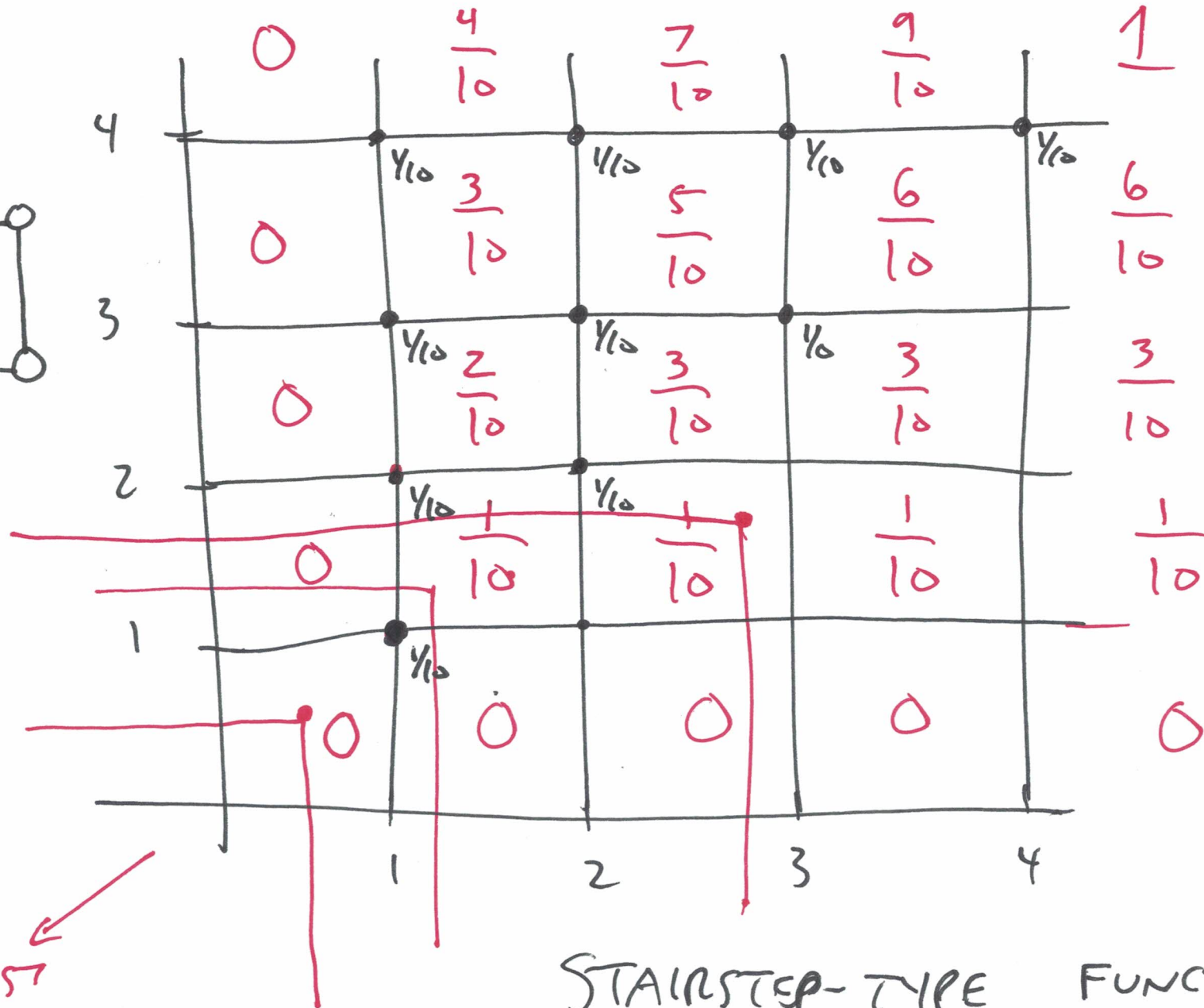
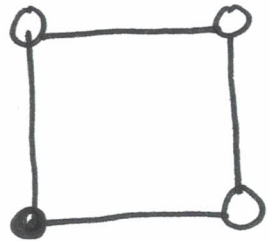
$$x \in \{1, 2, 3, 4\}$$

$$y \in \{x, \dots, 4\}$$



$$C = \frac{1}{10}$$

WHAT IS THE CDF?



STAIRSTEP-TYPE FUNCTION

DRAGON WITH 5 HP.

ROLL d6 TO DETERMINE ATTACK DAMAGE. (D)

THEN ROLL d4 TO DETERMINE BONUS

→ IF 3 OR 4, THEN ROLL ANOTHER

d6 (B). BUT DRAGON ALSO

GETS ADDITIONAL HEALTH OF $\left\lfloor \frac{B}{2} \right\rfloor$

↑
ROUND DOWN

$T = D + B$ TOTAL DAMAGE.

H = HIT POINTS OF DRAGON

$P_{TH}(x, y)$

SIDE TABLE:

BONUS	P(Bonus)	H	P(T B)											
			1	2	3	4	5	6	7	8	9	10	11	12
0	$\frac{1}{2}$	5	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$						
1	$\frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$	5		$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$					
2	$\frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$	6			$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$				
3	$\frac{1}{12}$	6				$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$			
4	$\frac{1}{12}$	7					$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$		
5	$\frac{1}{12}$	7						$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	
6	$\frac{1}{12}$	8							$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$\begin{aligned}
 &P(H=6, T=6) \leftarrow \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{12} \cdot \frac{1}{6} \\
 &= \cancel{P(H)} \cdot \cancel{P(\text{No Bonus}, D=6)} \\
 &\quad + \cancel{P(\text{Bonus}=1, D=5)}
 \end{aligned}$$

$$P(H=6, T=6)$$

$$\begin{aligned} &= P(\text{BONUS} = 2, D = 4) = \frac{1}{12} \cdot \frac{1}{6} \\ &\quad + P(\text{BONUS} = 3, D = 3) = \frac{1}{12} \cdot \frac{1}{6} \\ &\quad \hline &\quad \quad \quad \frac{2}{72}. \end{aligned}$$

$$T \Rightarrow P(T) = P(T|B)P(B)$$

	1	2	3	4	5	6	7	8	9	10	11	12
5	$\frac{6}{72}$	$\frac{1}{72}$ $\frac{1}{72}$	$\frac{1}{72}$	$\frac{1}{72}$	$\frac{1}{72}$	$\frac{1}{72}$	$\frac{1}{72}$					
6			$\frac{1}{72}$	$\frac{2}{72}$	$\frac{2}{72}$	$\frac{2}{72}$	$\frac{2}{72}$	$\frac{2}{72}$	$\frac{1}{72}$			
7				$\frac{1}{72}$ $\frac{1}{72}$	$\frac{1}{72}$	$\frac{2}{72}$	$\frac{2}{72}$	$\frac{2}{72}$	$\frac{2}{72}$	$\frac{2}{72}$	$\frac{1}{72}$	
8							$\frac{1}{72}$	$\frac{1}{72}$	$\frac{1}{72}$	$\frac{1}{72}$	$\frac{1}{72}$	$\frac{1}{72}$

$$35 + 7 + 12 + 12 + 6$$

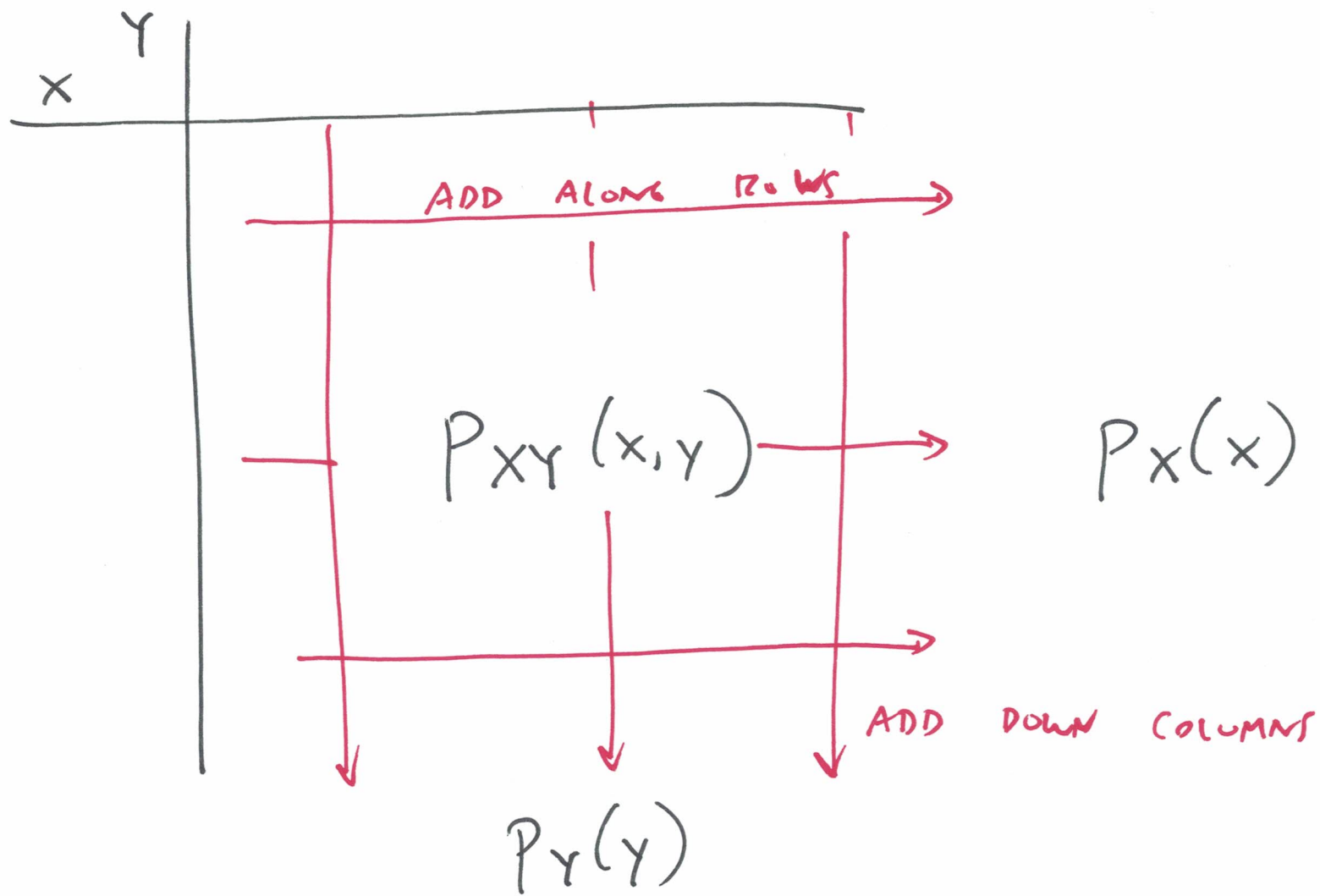
$$\frac{42}{24} = 72$$

$$P_{HT}(x, y)$$

KEY CONCEPT: MARGINAL PMF:

THE PMF FOR JUST ONE VARIABLE

IF WE DON'T CARE ABOUT THE OTHER.



MARGINAL PMF OF T IN EXAMPLE:

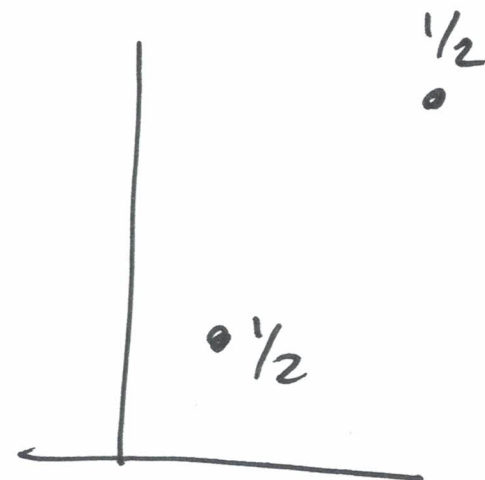
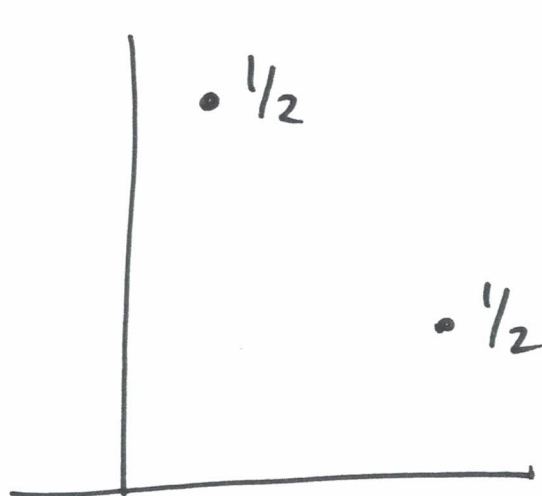
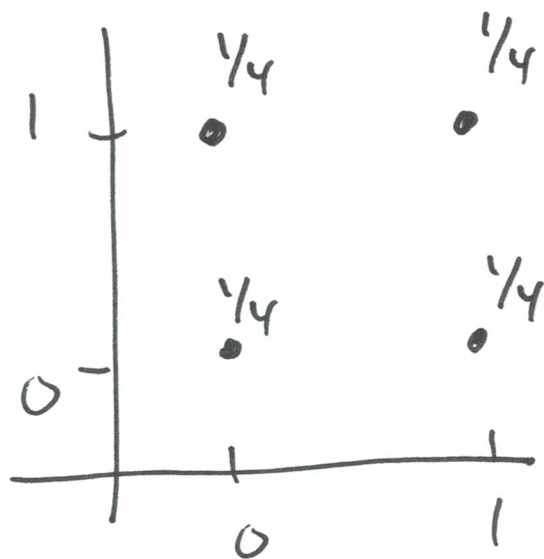
t	1	2	3	4	5	6	7	8	9	10	11	12
$P_T(t)$	$\frac{6}{72}$	$\frac{7}{72}$	$\frac{8}{72}$	$\frac{9}{72}$	$\frac{10}{72}$	$\frac{11}{72}$	$\frac{6}{72}$	$\frac{5}{72}$	$\frac{4}{72}$	$\frac{3}{72}$	$\frac{2}{72}$	$\frac{1}{72}$

MARGINAL PMF OF H IN EXAMPLE:

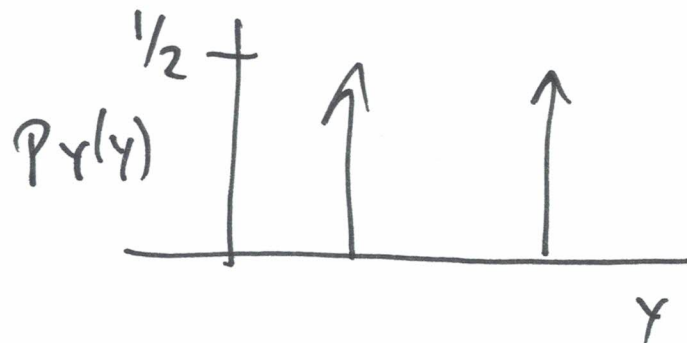
h	5	6	7	8
$P_H(h)$	$\frac{42}{72}$	$\frac{12}{72}$	$\frac{12}{72}$	$\frac{6}{72}$

EITHER WAY, ALL PROBABILITY IS
ACCOUNTED FOR (ADDS TO 1).

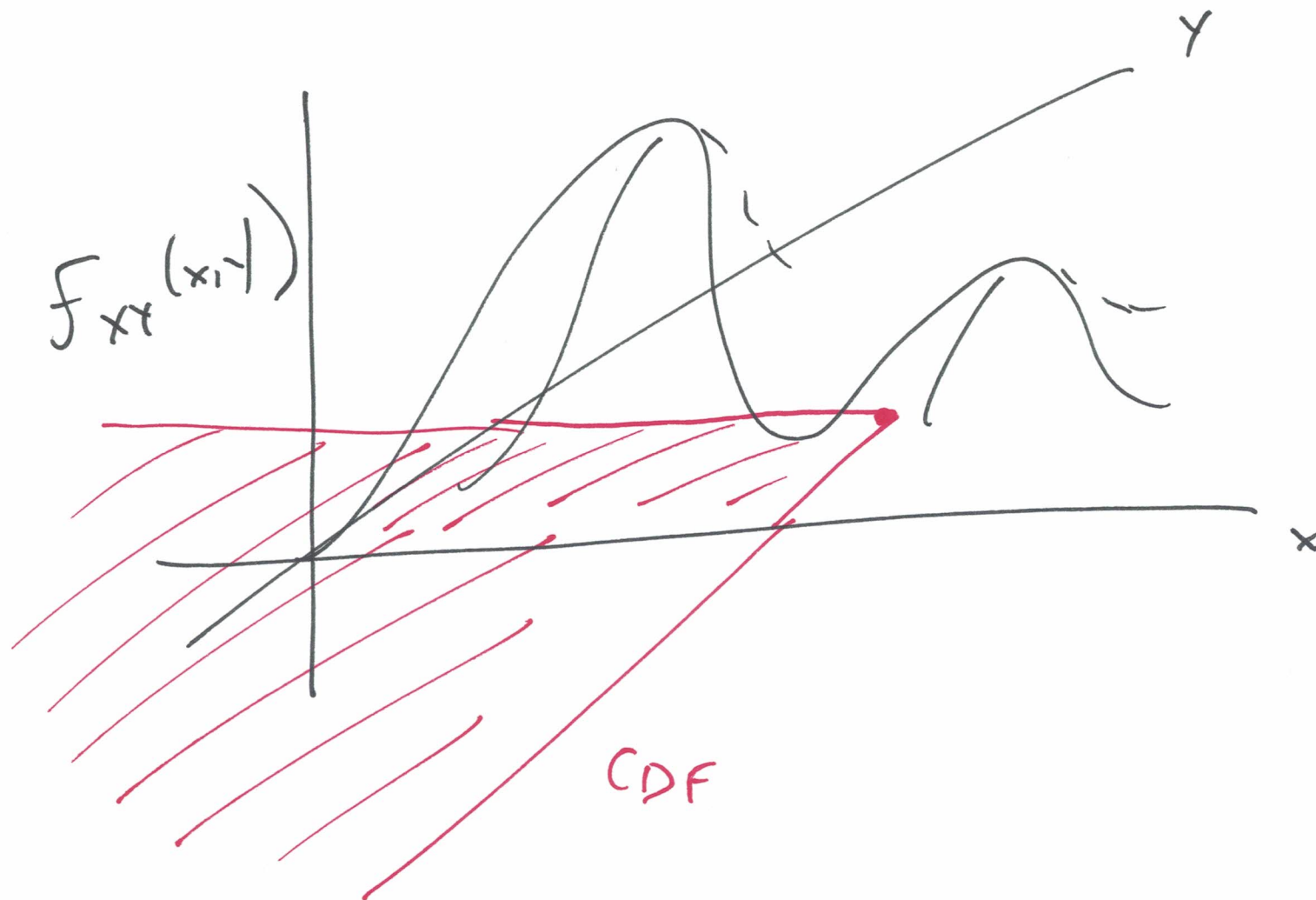
THE MARGINALS AREN'T ENOUGH TO
DETERMINE THE JOINT PMF!



ALL HAVE THE SAME MARGINALS



PREVIEW: CONTINUOUS PDF / CDF / MARGINALS:



PREVIEW (AFTER EXAM 2)

CONDITIONAL PMF/PDF

$$P_{X|Y}(x|y)$$